The Zeta Function of the T^4/Z_3 Torus:

Orbifold Projection, Dirichlet L-Functions and Sector Pairing of the GF(27) Classes

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Abstract

The Galois factorisation classes of Doc. 342 and the harmonic series on the T^4 torus of Doc. 159 together yield a precise statement about the zeta function of the FFGFT spacetime.

Four results: **(A)** The Z_3 orbifold projection of T^4/Z_3 transforms the full Riemann zeta into a Dirichlet L-function: $\zeta_{T^4/\mathbb{Z}_3}(s) = L(s, \chi_0) = (1 - 3^{-s})\zeta(s)$ **[B]**. **(B)** The Euler product of this L-function is hierarchically ordered by the entry depth $k = \text{ord}_p(3)$; the physically visible primes $\{2, 3, 5, 7, 11, 13\}$ are exactly those with $k \leq 6$ **[B]**. **(C)** The orbifold projection generates additional zeros at $s = 2\pi ik/\ln 3$ ($k \in \mathbb{Z} \setminus \{0\}$) on the imaginary axis $\text{Re}(s) = 0$; these are the algebraic signature of the Z_3 sector separation **[B]**. **(D)** The four Frobenius conjugacy classes of primitive elements in $\text{GF}(27)^*$ (Doc. 342, Theorem A) are paired into *two* classes by the FFGFT sector pairing $k \leftrightarrow -k$ (Doc. 336; on $\text{GF}(27)^*$: $x \mapsto x^{-1}$), $f_1 \leftrightarrow f_2$ and $f_3 \leftrightarrow f_4$; negation $x \mapsto -x = x^{13}\,x$ couples each primitive class to exactly one order-13 class **[B]**. The Galois character of $\text{Gal}(\text{GF}(27)/\text{GF}(3)) \cong \mathbb{Z}_3$ does not distinguish the classes; the distinction lies in $(\mathbb{Z}/26)^*/\langle 3\rangle \cong \mathbb{Z}_4$. Sector pairing and the Majorana criterion $k \mapsto k^{-1}$ (Doc. 340) partition the two layers identically: $\{f_3, f_4\}$ Dirac (charged leptons), $\{f_1, f_2\}$ Majorana (ν_1, ν_2 active; ν_3 in Dirac orbit O_4) **[B]**; prediction: ν_3 has no Majorana phase **[B]**.

Theorem D‴: quantum numbers of the second layer

What quantum numbers would the two layers carry? The algebra answers this without physical assumption (`pruef_343f_zweite_schicht.py`). Each primitive class f_i lies in exactly one $\mathbb{Z}_{13}$ orbit of the Frobenius (Doc. 339); with the exponents of Table 3: $f_1 = \{1, 3, 9\} \mapsto O_1$, $f_2 = \{17, 23, 25\} \mapsto O_3 = \{4, 10, 12\}$, $f_3 = \{5, 15, 19\} \mapsto O_2 = \{2, 5, 6\}$, $f_4 = \{7, 11, 21\} \mapsto O_4 = \{7, 8, 11\}$. Doc. 340 takes the self-inverseness of an orbit under $k \mapsto k^{-1} \pmod{13}$ as the Majorana criterion. Both involutions together give **[B]**:

Table 1: Quantum numbers of the four primitive classes from the algebra

Class	$\mathbb{Z}_{13}$ orbit	$k \mapsto -k$	$k \mapsto k^{-1}$	profile	layer
f_1	O_1	$\to f_2$	self-inverse	Majorana	$\{f_1, f_2\}$
f_2	O_3	$\to f_1$	self-inverse	Majorana	$\{f_1, f_2\}$
f_3	O_2	$\to f_4$	$\to O_4$	Dirac	$\{f_3, f_4\}$
f_4	O_4	$\to f_3$	$\to O_2$	Dirac	$\{f_3, f_4\}$

The sector pairing $k \mapsto -k$ and the Majorana criterion $k \mapsto k^{-1}$ partition the four classes identically: $\{f_1, f_2\}$ **is the Majorana layer,** $\{f_3, f_4\}$ **the Dirac layer [B]**. With the negation $f_i \leftrightarrow g_i$ the numbering of Doc. 342 is thus the natural one: sector pairing pairs adjacent indices, negation equal indices. Both layers carry $\mathbb{Z}_2$ parity 1 (massive sector, Doc. 339), no colour, and three states per class (three-cycle).

Interpretation [S]. Doc. 341 fixes order 26 in G(6) for the leptons; the class follows here: charged leptons are Dirac particles and belong to $\{f_3, f_4\}$. The second layer is thus not new but the neutrino sector: Doc. 340 places the active neutrinos in $O_3 = f_2$. What remains open is $f_1 = O_1 = \{1, 3, 9\}$ — the orbit of the generator itself: Majorana, three states, $\mathbb{Z}_2$ parity 1, sector partner of the active neutrinos under $k \mapsto -k$. That is the profile of sterile (right-handed) neutrinos, which Doc. 340 already predicts as Majorana in the orbit-3 sector with $m_{\text{min}} = 4\, m_\nu = 18.2$ meV; the pairing $O_1 \leftrightarrow O_3$ under the sector pairing is algebraically the seesaw pairing active↔sterile. A curiosity without status: $(\max O_1)^2 - 1 = 80 = |\text{GF}(81)^*|$, analogous to $(\max O_4)^2 - 1 = 120 = \Delta m^2_{\text{atm}}/m^2_\nu$ from Doc. 340. Whether O_1 carries the mass scale of the sterile neutrinos is open.

Two further results: **(E)** The harmonic primes $\{5, 7, 11, 13\}$ are uniquely forced by the prime factorisations of $3^k - 1$ for $k = 3, 4, 5, 6$; $p = 13$ is the worldwide unique prime with $\text{ord}_p(3) = 3$ **[B]**. **(F)** The Euler factor threshold at $p^* \approx 9.76$ ($s = 2$) marks the 7-limit exactly; from $p_{\text{max}} \approx 30$ the Galois product density lies at chance level; the Weyl obstruction (Doc. 316) shows that the limit lies in the object; Shor's circuit model is digital and unaffected, but the physical apparatus is a resonant system subject to the resonance limits as long as the threshold theorem is not demonstrated at scale **[S]**; empirically no quantum advantage has been demonstrated [X].

Check scripts: `pruef_343_zeta_galois.py`, `pruef_343b_klassen_symmetrie.py`, `pruef_344_primzahlmuster.py`.

.1 Status markers and starting point

The following status markers are used throughout:

[K] *Confirmed* — fully proved from ξ or a corpus-established derivation; all check-script assertions passed.
[B] *Established* — numerically secured; analytical proof present or trivial.
[S] *Speculative* — indication or conjecture, not yet fully proved.

Doc. 159 shows that the spectral zeta function of the Laplace operator on a torus T^d is the Epstein zeta function, whose one-dimensional special case is the Riemann zeta $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$. Doc. 342 shows that primes appear in the GF(3^k) hierarchy of the T^4/Z_3 orbifold ordered by entry depth $k = \mathrm{ord}_p(3)$. This document combines both: what does the GF(3^k) hierarchy imply for the zeta function itself?

.2 Theorem A: Orbifold projection and Dirichlet L-function

Decomposition of the zeta by Z_3 classes

On T^4/Z_3 the allowed modes $n \in \mathbb{Z}^4$ are structured by Z_3 invariance. In the one-dimensional analogue $\zeta(s)$ splits into three partial sums by n mod 3 **[B]**:

$$\zeta(s) = \underbrace{\sum_{\substack{n=1\\3|n}}^{\infty} \frac{1}{n^s}}_{=3^{-s}\zeta(s)} + \underbrace{\sum_{\substack{n=1\\n\equiv1(3)}}^{\infty} \frac{1}{n^s}}_{L_1(s)} + \underbrace{\sum_{\substack{n=1\\n\equiv2(3)}}^{\infty} \frac{1}{n^s}}_{L_2(s)} \tag{1}$$

The fixed-point sector ($n \equiv 0$) contributes exactly $3^{-s}\zeta(s)$; the twist sectors ($n \equiv \pm 1$) together $(1-3^{-s})\zeta(s)$.

Principal character χ_0 mod 3

The Dirichlet character χ_0 mod 3 with $\chi_0(n) = 0$ if $3 \mid n$, otherwise $\chi_0(n) = 1$, gives **[B]**:

$$L(s,\chi_0) = \sum_{\mathrm{gcd}(n,3)=1} \frac{1}{n^s} = (1-3^{-s})\,\zeta(s) = \prod_{p\neq 3} \frac{1}{1-p^{-s}} \tag{2}$$

This is the **spectral zeta function of the T^4/Z_3 torus**: the characteristic prime $p = 3$ does not appear as an Euler factor — it is the geometric foundation of the orbifold and stands outside the prime hierarchy.

Non-trivial character χ_1 mod 3

The character χ_1 mod 3 with $\chi_1(1) = +1$, $\chi_1(2) = -1$, $\chi_1(0) = 0$ measures the *asymmetry* of the twist sectors **[B]**:

$$L(s,\chi_1) = \sum_{\substack{n=1\\n\equiv1(3)}}^{\infty} \frac{1}{n^s} - \sum_{\substack{n=1\\n\equiv2(3)}}^{\infty} \frac{1}{n^s} \tag{3}$$

with the exact value (Dirichlet 1837): $L(1,\chi_1) = \pi/(3\sqrt{3})$.

Physical meaning **[S]**: $L(s,\chi_1)$ measures the difference between winding numbers $w=+1$ and $w=-1$ on the torus, i.e. the Z_3 sector asymmetry. Whether $L(s,\chi_1)$ has a direct physical observable in FFGFT (particle–antiparticle asymmetry in the twisted sector?) remains open **[S]**.

.3 Theorem B: Hierarchical Euler product by $k = \mathrm{ord}_p(3)$

The primes in $L(s,\chi_0) = \prod_{p\neq 3}(1-p^{-s})^{-1}$ are not equivalent. By the entry-depth rule $k = \mathrm{ord}_p(3)$ from Doc. 342, Theorem B, the Euler product is hierarchically ordered **[B]**:

$$L(s,\chi_0) = \prod_{k=1}^{\infty} \prod_{\substack{p\neq 3 \\ \mathrm{ord}_p(3)=k}} \frac{1}{1-p^{-s}} \tag{4}$$

Table 2: Hierarchy of Euler factors by $k = \mathrm{ord}_p(3)$

k	new prime(s)	Euler factor at $s=2$	harmonics	FFGFT role
1	2	$4/3 = 1.333$	octave	Z_2 fixed point, massive sector
3	13	≈ 1.006	thirteenth	GF$(27)^*$, leptons
4	5	≈ 1.042	major third	factor in $(m_\mu/m_e)^2$
5	11	≈ 1.008	eleventh	neutrinos (Doc. 340)
6	7	≈ 1.021	seventh	top quark (Doc. 289)
≥ 7	$\geq 41, 1093, \ldots$	$\approx 1 + p^{-2}$	no small harmonics	suppressed

Fractal suppression of higher Euler factors

On the fractal T^4 torus (Doc. 159) high-mode contributions are suppressed by the damping factor $\tau^{-(1+\xi)k}$. This transfers to the Euler factors; the *weighted* Euler product reads

$$\zeta^{\text{frac}}_{T^4/\mathbb{Z}_3}(s) \;\cong\; \prod_{k=1}^{\infty} \prod_{\substack{p\neq 3 \\ \mathrm{ord}_p(3)=k}} \left(\frac{1}{1-p^{-s}}\right)^{\tau^{-k(1+\xi)}} \tag{5}$$

Since $\tau^{-k(1+\xi)} \to 0$ as $k\to\infty$, each Euler factor is pushed toward 1 for large k: primes with $\mathrm{ord}_p(3) \geq 7$ make no physical contribution **[B]**. This is Theorem D of Doc. 342 in the language of the zeta function.

.4 Theorem C: New zeros from the orbifold projection

Zeros of $(1-3^{-s})$

The factor $(1-3^{-s})$ in $L(s,\chi_0) = (1-3^{-s})\zeta(s)$ has zeros at **[B]**:

$$3^{-s} = 1 \quad\Longleftrightarrow\quad s = \frac{2\pi i\, k}{\ln 3}, \qquad k \in \mathbb{Z}\setminus\{0\} \tag{6}$$

These lie on $\mathrm{Re}(s) = 0$ (imaginary axis), *not* on the critical line $\mathrm{Re}(s) = 1/2$ of the Riemann hypothesis. They are absent from the Riemann zeta spectrum; they arise solely from the Z_3 orbifold projection.

The zero spacing $2\pi/\ln 3 \approx 5.72$ encodes the periodicity of the Z_3 lattice structure. A resonance test of the first eight known Riemann zeros γ_n against this unit shows no significant coincidence (deviations 0.15–0.47) **[B]**: the full Riemann zeta encodes the distribution of *all* primes, not only the Z_3-compatible ones.

Complete zero structure of the orbifold zeta

$L(s,\chi_0)$ has three types of zeros:

- **Trivial zeros of** $\zeta(s)$: at $s = -2, -4, -6, \dots$
- **Non-trivial Riemann zeros**: on $\mathrm{Re}(s) = 1/2$ (assumed), unchanged
- **Orbifold zeros**: at $s = 2\pi i k/\ln 3$, $k \neq 0$ — new, generated by the Z_3 projection **[B]**

.5 Theorem D: The four classes under sector pairing

The Galois character does not distinguish the classes

The Galois group $\mathrm{Gal}(\mathrm{GF}(27)/\mathrm{GF}(3)) = \langle F\rangle \cong \mathbb{Z}_3$ with $F\colon\ x \mapsto x^3$ has three characters $\chi_j(F) = \omega^j$, $j = 0, 1, 2$. All four primitive classes $f_1, \dots, f_4$ of Doc. 342 are *regular* F-orbits of length 3: F acts on each as a 3-cycle, and the associated character is the same for all four. The Galois character therefore does not separate the classes **[B]**.

The distinction lies in the discrete logarithm: with a generator x of $\mathrm{GF}(27)^*$, each class is an orbit $\{x^k, x^{3k}, x^{9k}\}$ with $\gcd(k, 26) = 1$, i.e. a coset of $\langle 3\rangle = \{1, 3, 9\}$ in $(\mathbb{Z}/26)^* \cong \mathbb{Z}_{12}$. There are $12/3 = 4$ cosets **[B]**:

Table 3: The four primitive classes as cosets of $\langle 3\rangle$ in $(\mathbb{Z}/26)^*$ (basis $x^3 + 2x + 1 = 0$)

Class	Minimal polynomial	Exponents k	Role
f_1	$x^3 + 2x + 1$	$\{1, 3, 9\}$	Majorana layer (Theorem D‴)
f_2	$x^3 + 2x^2 + 1$	$\{17, 23, 25\}$	$= f_1^{-1}$, Majorana layer
f_3	$x^3 + 2x^2 + x + 1$	$\{5, 15, 19\}$	Dirac layer
f_4	$x^3 + x^2 + 2x + 1$	$\{7, 11, 21\}$	$= f_3^{-1}$, Dirac layer

The quotient $(\mathbb{Z}/26)^*/\langle 3\rangle \cong \mathbb{Z}_4$ is cyclic; inversion $k \mapsto -k$ is its element of order 2.

Sector pairing reduces four to two

Doc. 336 identifies the FFGFT sector pairing $k \leftrightarrow -k$ with the Frobenius automorphism; at the level of elements of $\mathrm{GF}(27)^*$ it is inversion $x \mapsto x^{-1} = x^{25}$. It permutes the classes as **[B]**

$$f_1 \longleftrightarrow f_2, \qquad f_3 \longleftrightarrow f_4. \tag{7}$$

If sector pairing is a symmetry of the mass layer — as it is in the corpus — there are not four but **two** physically distinguishable $\mathrm{GF}(27)^*$ layers: $\{f_1, f_2\}$ and $\{f_3, f_4\}$ (assignment in Theorem D‴).

Negation couples order 26 to order 13

Since $-1 = x^{13}$ in GF(27), the map $x \mapsto -x$ sends every element of order 26 to one of order 13 and vice versa. The four order-13 classes $g_1, \dots, g_4$ of Doc. 342 are therefore not an independent structure but the sign partners of the primitive classes **[B]**:

$$f_i \leftrightarrow g_i \qquad (i = 1, \dots, 4). \tag{8}$$

The candidate "sub-lepton structure" from the order-13 polynomials, left open in Doc. 342, is thereby eliminated.

Open question

The reduction $4 \to 2$ sharpens the question of Doc. 342: not "do f_2, f_3, f_4 carry further layers", but "there are exactly two GF$(27)^*$ layers $\{f_1, f_2\}$ and $\{f_3, f_4\}$ — which particles carry which?" Theorem D‴ answers this from the algebra.

.6 Theorem E: Prime patterns in the GF(3^k) hierarchy

The harmonic primes are uniquely forced

A prime p satisfies ord$_p(3) = k$ if and only if p divides $3^k - 1$ and $p \nmid 3^j - 1$ for all $j < k$. The prime factorisations of the first relevant values give **[B]**:

Table 4: Prime factors of $3^k - 1$ and the forced physical primes

k	$3^k - 1$	prime factors	phys. prime	note
1	2	$\{2\}$	2	octave
3	26	$\{2, 13\}$	**13**	unique prime with ord$_p(3) = 3$ worldwide
4	80	$\{2, 5\}$	**5**	only non-2 factor
5	242	$\{2, 11\}$	**11**	only non-2 factor
6	728	$\{2, 7, 13\}$	**7**	(13 has ord$_{13}(3) = 3$, not 6)

The strongest result concerns $p = 13$ **[B]**: since $3^3 - 1 = 26 = 2 \cdot 13$ and ord$_2(3) = 1$, $p = 13$ is the *unique* prime worldwide with ord$_p(3) = 3$. There is no other, and there can be none. The physical primes $\{5, 7, 11, 13\}$ are not selected by harmonic considerations, but are completely and uniquely forced by the prime-factor structure of $3^k - 1$ for $k = 3, 4, 5, 6$.

Necessary congruence condition

From ord$_p(3) = k$ with $3 \mid k$ it follows that $3 \mid (p - 1)$, i.e. $p \equiv 1 \pmod 3$ **[B]**. In particular $p = 13$ (at $k = 3$) and $p = 7$ (at $k = 6$) are both $\equiv 1 \pmod 3$ — exactly the Z$_3$-compatible primes of the corpus (Doc. 342, Theorem B).

.7 Theorem F: Limits of distinguishability and computational cost

Exact threshold: Euler factor vs. fractal correction

The Euler factor $1/(1 - p^{-s})$ exceeds the fractal correction $1.05\,\%$ (Doc. 338, bare to measured) exactly for $p < p^*$ with $p^* = (0.0105)^{-1/s}$ **[B]**:

Table 5: Exact threshold p^* by value of s

s	p^*	primes $\leq p^*$	harmonic limit	Euler factor $\geq 1.05\,\%$
1	95.2	$\{2, 3, 5, \ldots, 89\}$	—	all small primes
2	9.76	$\{2, 3, 5, 7\}$	7-limit	octave, fifth, third, seventh
3	4.57	$\{2, 3\}$	3-limit	octave and fifth only

For $s = 2$ (reference value of the spectral zeta) the threshold is $p^* \approx 9.76$: primes $\{2, 5, 7\}$ have Euler factors exceeding the fractal correction; $p = 11$ and all larger primes fall below it. This is exactly the harmonic 7-limit. $p = 11$ appears in the corpus (neutrinos, Doc. 340) via the structural formula $\Delta m^2 = (11^2 - 1)\, m_\nu^2$, not as a Galois product hit.

Galois products vs. random products: hit rate

Forming all products of up to three Galois building blocks with exponents $\{-1, 1, 2\}$ for $p \leq p_{\text{max}}$, the product set grows rapidly and the hit rate for arbitrary targets reaches chance level **[B]**:

Table 6: Hit rate for physical vs. random targets at $\pm 0.5\,\%$

p_{max}	values	43200	m_μ/m_e	random avg
13	1334	2	0	0.92
30	5802	3	2	3.60
50	8946	3	2	5.33
100	13398	4	3	7.66
300	24992	8	12	14.15

From $p_{\text{max}} \approx 30$ the hit rate lies at chance level ($\pm 0.5\,\%$); at the strict threshold $\pm 0.05\,\%$ already from $p_{\text{max}} \approx 13$. The identity $43200 = |\text{GF}(9)^*|^2 \cdot 5^2 \cdot |\text{GF}(27)|$ (Doc. 338) uses only building blocks with $p \leq 5$ and is derived *geometrically* from ξ, not found by product search.

Structural computational limit: Weyl obstruction and quantum computers

The distinguishability threshold at $p_{\text{max}} \approx 13$ is a *physical* limit (fractal correction). Independently of this, the corpus contains a sharper *mathematical* limit that applies to every technology **[B]**:

Weyl obstruction, Doc. 316

There is no compact Riemannian manifold — of any dimension, curvature, topology, or orbifold structure — whose Laplace spectrum consists of the Riemann zeros. Reason: $T \ln T$ is not a power $T^{d/2}$.

From Doc. 147, an n-qubit register is a compact state space with eigenvalue density $N(\lambda) \sim \lambda^{d/2}$; arithmetic counting densities grow as $T \ln T$. A finite state space cannot fully carry such a spectrum, *independently of the register size* **[B]**.

Doc. 075 names the consequence for Shor's algorithm:

Method	Limit	Type
Trial division	$O(\sqrt{n})$	computational
Pollard ρ	$O(n^{1/4})$	computational
Shor's QFT	resource limit	Weyl obstruction

The central statement of Doc. 316 holds technology-independently:

> *The limit lies not in the tool but in the object.* Transformation (Fourier, QFT, optics) projects and generates no information. Relaxation finds what is already in the potential. Both presuppose that the sought structure is already present.

Physical limit ($p^* \approx 9.76$, Theorem F.1) and mathematical limit (Weyl obstruction, Theorem F.3) thus appear in the same picture: beyond a certain point the prime structure is hidden in the noise, and no analogue tool retrieves it; Shor's algorithm is digital as a model, a resonator as an apparatus — and thus, until proven otherwise, bound as well (Theorem G′).

.8 In plain terms: what the limit means

Question. So arbitrarily dense harmonics can be computed — but beyond a certain depth they can no longer be distinguished?

Answer. Yes, exactly that.

Computation can go arbitrarily far. GF(3^k) for $k = 7, 8, 12, 100$ — the algebra does not stop. Every k yields new primes, new polynomial classes, new chords. At $k = 12$ all four harmonic primes $5, 7, 11, 13$ sit in one field simultaneously; at $k = 7$ the Wieferich prime 1093 enters. These structures are exactly computable and mathematically all present.

But beyond a certain depth they are physically meaningless, for two reasons that coincide.

First, the grid becomes too fine. For $k \leq 3$ the reachable values lie about $3\,\%$ apart — if 43200 hits a mass value to $1\,\%$, that is a result. For $k \geq 5$ they lie less than $0.5\,\%$ apart. Then *every* number hits something, and a hit means nothing. The control test in Theorem F shows it: two hundred random numbers hit as often as the actual mass ratios.

Second — and this is the physical reason — the fractal correction of about $1\,\%$ (Doc. 338, bare to measured) is itself already a statement about higher winding modes. Once the Galois grid is finer than this correction, nature itself no longer distinguishes neighbouring Galois values. The structure is not gone; it has dissolved into the noise of the higher modes.

The analogy with harmonics is exact: everyone hears the overtones $2:1$ and $3:2$. $17:16$ is a semitone, $33:32$ a quarter-tone. From roughly the 250th overtone the spacing falls below 5 cents — mathematically exactly present, acoustically no longer distinguishable. The series becomes a continuum.

The consequence for FFGFT. The exact Galois layer reaches to GF(27), i.e. $k = 3$: the lepton layer, 43200, the four classes, $p = 13$ as the unique prime with $\mathrm{ord}_p(3) = 3$. Above that ($k = 4$, GF(81), $p = 5$) is already the transition. From $k = 5$ on, anything found by search is indistinguishable from chance — and must receive no status unless it comes directly from ξ.

This is not a deficiency of the theory but its consistency: Doc. 159 says mode convergence, Doc. 316 says Weyl obstruction, Doc. 342 and this document say the same thing algebraically. There is a limit, it lies at $k \approx 3$–4, and it comes from the subject itself, not from our computational capacity.

.9 In plain terms II: synthesis, analysis, factorisation — and coupling

Question. What does the limit mean for synthesis and analysis — and for factorisation? And how does coupling enter: Doc. 328 has three coupling regimes; do pure numbers show the same picture as physical processes?

Synthesis is unlimited, analysis is limited

Synthesis — adding overtones, multiplying primes, forming GF(3^k) chords from partitions — goes arbitrarily far and arbitrarily dense. Nothing prevents superposing the 250th overtone with the 251st or multiplying two 300-digit primes. Synthesis does not know the limit of Theorem F.

Analysis — which overtones are in the sound, which primes in the number, which Galois class carries the mass — hits exactly this limit. It is not a matter of computational cost but of resolution: to separate overtone n from $n+1$ one needs a time window of about n periods; to find the period r of a number, a frequency resolution $\Delta f \approx f_0/n^2$ (Doc. 075, precision barrier). Beyond a certain density the spacing lies below the noise, and the decomposition is no longer unique — not because it does not exist, but because many decompositions fit equally well. That is the chance-level finding of Theorem F in other words.

Factorisation is analysis

Factorising $N = p \cdot q$ means reading the two fundamentals out of a beat. For small p, q — the 7-limit range, $k \leq 3$ — this works; those are exactly the primes visible in the FFGFT spectrum at all. RSA primes with 300 digits sit at $k = \text{ord}_p(3)$ of the order of p itself, i.e. at a depth undercutting the grid spacing of Theorem F by hundreds of orders of magnitude. Doc. 316 puts it thus: Fourier, QFT and optics project, they generate no information; the period must already be in the prepared states, and this preparation (modular exponentiation, $95\,\%$ of the gate cost, Doc. 147) is exactly the problem one wanted to solve.

Two things follow. First: FFGFT does not break RSA — and that is not a deficiency but a consistent result. The asymmetry synthesis-easy / analysis-hard on which RSA lives is the same asymmetry that ends the Galois layer at $k \approx 3$–4. Second, for instrument building: an instrument can *produce* overtones arbitrarily densely; ear and analysis *hear* them sharply only up to about the 7-limit. What lies above acts as timbre, not as pitch — as the higher winding modes in FFGFT act as a $1\,\%$ correction, not as a separate mass layer.

Caveat [S]. Transferring Theorem F to $\mathbb{Z}/N\mathbb{Z}$ is an analogy. Doc. 075 records that ξ has no function in $\mathbb{Z}/N\mathbb{Z}$ and the period is fixed purely arithmetically by $\text{lcm}(p-1, q-1)$. The analogy holds; the fractal correction as the *mechanism* of the factorisation limit does not.

Coupling: the same threefold division, not the same law

Doc. 328 classifies couplings relative to loss rates: subcritical ($\kappa < 1$, one maximum), critical ($\kappa = 1$, maximally flat), supercritical ($\kappa > 1$, splitting). Locking onto a frequency ratio p/q occurs in Arnold tongues of width $\Delta\Omega(p/q) \propto K^q$.

For pure numbers the tolerance ε takes the role of the coupling strength: a real number "locks" onto p/q if $|x - p/q| < \varepsilon$. Fractions with denominator $\leq Q$ cover the interval with measure $\approx 2\varepsilon \cdot 3Q^2/\pi^2$; coverage 1 gives the critical denominator height **[B]**

$$Q_c = \frac{\pi}{\sqrt{6\,\varepsilon}}. \tag{9}$$

For $\varepsilon = 1.05\,\%$ (fractal correction), $Q_c \approx 12.5$ — the same order as $p_{\text{max}} \approx 13$ in Theorem F. The Galois version confirms it numerically (`pruef_343c`): at $\varepsilon = 1.05\,\%$ the Galois grid with $p_{\text{max}} = 7$ is subcritical (coverage 0.52), with $p_{\text{max}} = 13$ critical (0.92), from $p_{\text{max}} \approx 30$ supercritical (1.00). **The limit $k \approx 3$–4 of Theorem F is the critical point $\kappa = 1$ in the sense of Doc. 328 [B]**.

So the threefold division is the same in arithmetic and physics. *The law behind it is not* **[B]**: Arnold tongues fall exponentially in q ($\Delta\Omega \propto K^q$), Farey spacings only polynomially ($1/q^2$). At $K = 0.3$ and $q = 5$ the tongue is already 10^{-5} narrower than the arithmetic capture interval; at $q = 13$ by 10^{-15}. Physical locking is exponentially selective, arithmetic distinguishability polynomial. Physics "hears" fewer harmonics than arithmetic permits — the edge is the same, but physically sharper.

FFGFT is dynamically deeply subcritical

With $K_{\text{eff}} = 2\pi\xi \approx 8.4 \times 10^{-4}$ (Doc. 328) the widest non-trivial Arnold tongue is $\Delta\Omega(1/2) \approx 5.6 \times 10^{-8}$, the total measure of all tongues $\approx 8 \times 10^{-4}$. Both lie five to two orders of magnitude below the fractal correction **[B]**. No mass ratio is pulled onto a p/q by resonant locking.

This yields a clarification that was so far only implicit in the corpus: the exact rationals — 43200, $8^2 \cdot 5^2 \cdot 27$, the winding numbers (r_i, p_i) — are *topological* (integer windings on T^4), not *dynamical* (locking). Discreteness and coupling are two separate mechanisms in FFGFT. This is consistent with Doc. 328, where φ lies outside all tongues: the coupling is too weak to hold ratios; what is held, topology holds.

.10 Theorem G′: ξ as resolution floor — the limit depends on the level

Question. Is it excluded that ξ plays a role in resonance finding or analysis?

Answer. Not absolutely. Three roles must be distinguished. As a *selection criterion* in a search function of the form $\exp(-(\omega - \omega_0)^2/4\xi)$, ξ is excluded: the centre ω_0 does the work, ξ only sets the width **[B]**. As a *dynamical coupling*, ξ is excluded: $K_{\text{eff}} = 2\pi\xi$ gives Arnold tongues of 5.6×10^{-8} (Theorem G″) **[B]**. As a *resolution floor*, however, ξ is the natural scale: Doc. 328 sets $g/\omega = \xi$ as the linewidth of the torus modes, Doc. 162 the fractal damping per operation as $K_{\text{frak}} = 1 - 100\xi$.

The 1.05 % is one fractal level

The correction bare → measured of Doc. 338 ($1.05\,\%$) and the damping level $100\xi = 1.33\,\%$ of Doc. 162 stand in the ratio 0.79 **[B]**. The thresholds in Theorems F and G were computed with $\varepsilon = 1.05\,\%$ — they are thus the thresholds of *one* fractal level, not of the theory. If this level is subtracted analytically (it is computable per Docs. 338/162), the resolving power moves to ξ:

Table 7: Thresholds of Theorems F/G for the corpus resolution levels (`pruef_343d_xi_aufloesung.py`)

Resolution ε	value	Q_c	p^*	primes $\leq p^*$	Galois resolved to
$1.05\,\%$ (Doc. 338)	1.05×10^{-2}	12.5	9.8	4	$k = 3$
100ξ (Doc. 162)	1.33×10^{-2}	11.1	8.7	4	$k = 3$
ξ (Doc. 328)	1.33×10^{-4}	111	87	23	$k \geq 8$
$1/Q_{\text{TFLN}}$ (Doc. 186)	10^{-6}	1283	1000	168	$k \geq 8$
ξ^2	1.78×10^{-8}	9619	7500	950	$k \geq 8$

At $k_{\text{max}} = 8$ the Galois grid still has a mean spacing of $0.128\,\%$ — ten times coarser than $\xi = 0.0133\,\%$. At ξ resolution the layer remains resolved at least to $k = 8$, and primes up to 83 would have distinguishable Euler factors **[B]**.

Photonics as access to the ξ floor

Doc. 186 gives $Q > 10^6$ for TFLN resonators. Resolving ξ requires only $Q > 1/\xi = 7500$; resolving ξ^2 would require $Q > 5.6 \times 10^7$ **[B]**. Photonic hardware thus reaches down to the ξ floor, not below — whereas NISQ hardware, per Doc. 034, loses the ξ corrections in noise. This is the only place where ξ "plays a role in analysis": not as the tool that finds the resonance, but as the scale down to which anything can be distinguished at all. ξ has the same role everywhere in the corpus: $43200 = (r_\mu/r_e)^2/\xi$ says *where* the resonance is, not how to find it.

Caveats [S]. Whether the 100ξ level is actually subtractable is asserted in the corpus, not carried out. Doc. 186 gives $Q > 10^6$ for the optical phase, not for mass ratios; the transfer $1/Q \to \varepsilon$ is an analogy. The check of Doc. 186 itself (`pruef_343e_dok186_check.py`) confirms the TFLN quality but refutes its "locking" onto φ^{-k} (contradiction to Doc. 328 [K]), the Fibonacci condition for winding numbers ($r_\mu = 16/5$, $16 \notin F$) and the B-meson interpretation ($\alpha = \xi$ would need 9×10^8 events for 4σ); Doc. 186 carries a corresponding preliminary note since 3 September 2026.

Consequence for factorisation: $N_{\text{max}} \approx 1/\varepsilon$

A resonance-based factoriser must separate the period r from $r{+}1$ (resolution $1/r$), or, by the precision barrier of Doc. 075 ($\Delta f \approx f_0/n^2$), isolate the factor n (resolution $1/n^2$). The resolution floor ε therefore caps it at $N_{\text{max}} \approx 1/\varepsilon$ **[B]**:

Table 8: Resolution floor and ceiling of a resonance-based factoriser

ε	factors up to n_{max}	numbers up to N_{max}	bits
$1.05\,\%$ (one fractal level)	10	95	7
ξ	87	7.5×10^3	13
$1/Q_{\text{TFLN}}$	10^3	10^6	20
ξ^2	7.5×10^3	5.6×10^7	26
RSA-2048	—	would need $\varepsilon \approx 2^{-1024}$	—

The $p^* \approx 87$ of the threshold table *is* the factorisation statement: a resonance method at the ξ floor separates factors up to about 87. Even with ξ^2 that is 26 bits — trial division does it in milliseconds. What moves with the ξ floor is the Galois layer of physics ($k \approx 3 \to k \geq 8$), not factorisation: physics gains resolving power for mass ratios; number theory gains nothing, because its numbers carry no ξ scale (Doc. 075).

Why classical methods reach further: state versus process

Classical methods resolve nothing — they count. $a^r \bmod N = 1$ is true or false; there is no ε. Every bit is a $\mathbb{Z}_2$ state, in the corpus a winding number on T^4 (Doc. 247): a *state* that exists without energy and is re-thresholded to 0 or 1 after every step. That is the topological discreteness of Theorem G″. A resonance method, by contrast, carries its information as a *process* — phase, frequency — and has no state to threshold back to; the floor ε stays with it. The price of exactness is enumeration:

Table 9: Classical limits are counting limits (operations), not resolution limits

bits	trial division $\sqrt{N}$	Pollard-ρ $N^{1/4}$	NFS	
13	90	10	9×10^2	resonance ceiling at ξ
26	8×10^3	90	3×10^4	resonance ceiling at ξ^2
512	10^{77}	3×10^{38}	2×10^{19}	hours
2048	—	3×10^{154}	1.5×10^{35}	5×10^{12} years at 10^{15} op/s

On energy: Landauer's bound $k_B T \ln 2$ holds in the corpus unchanged and without ξ modification (Doc. 184), but *only for the erasure* of a bit (Doc. 247); Doc. 255 derives it as the frozen energy of a minimal winding difference $\Delta w = 1$ from T^4 geometry. It caps dissipation per irreversible step, not the number of steps; reversible steps are exempt. The classical wall is therefore time, not energy **[B]**.

The two method types are dual: **exactness costs enumeration, resolution costs precision.** Resonance has constant cost per candidate and a floor at $1/\varepsilon$; counting has no floor and a cost growing subexponentially in N. Both are limits; the counting limit merely lies hundreds of orders of magnitude further out. Doc. 316 says it generally: the arithmetic structure cannot be projected into a continuum without loss of information.

Quantum computers: the empirical yardstick

Two levels must be separated. The *circuit model* of Shor's algorithm is digital and untouched by the Weyl obstruction (Theorem F): the QFT runs over $\mathbb{Z}_Q$, $Q = 2^{2n}$, with equally spaced eigenfrequencies and linear counting density $N(\lambda) \propto \lambda$; the number theory sits in the period r of $a^x \bmod N$, not in the spectrum **[B]**. That holds on paper.

The *physical apparatus* is something else. A qubit is a two-level resonator, every gate a resonant drive (Rabi rotation), a photon a mode with continuous phase. Quantum computers and photonics do not compute digitally and do not enumerate; they are resonant systems and are subject to the resonance limits: the resolution floor ε (Theorem G′) and, wherever the arithmetic is to be carried in a spectrum, the Weyl obstruction. That resonators become "digital" is the claim of the threshold theorem (re-thresholding by projective measurement at gate error $< 10^{-3}$, approximate QFT with phase resolution $1/\mathrm{poly}(n)$, connectivity, magic-state distillation, modular exponentiation at $95\,\%$ of the gates, Doc. 147); each condition has been demonstrated individually, the product at scale has not. Until it is, the apparatus obeys what every resonator obeys. The substrate is immaterial — photonics, ions, superconductors differ only in the error channel (for TFLN, photon loss: < 1 dB per facet per Doc. 186 is up to $20\,\%$ per transition, above the loss thresholds of fusion-based schemes). The photonic operations of Doc. 186 (analogue matrix convolution, resonator locking) are analogue in any case and sit, with $\varepsilon = 1/Q \approx 10^{-6}$, at $N_{\max} \approx 10^6$ (20 bit).

There are thus several limits — resolution, Weyl, resources — and the corpus has no mechanism that frees a physical resonator from them (Doc. 034: ξ effects below noise; Doc. 162: 100ξ damping per gate, character open). Doc. 075 denies the obstruction for Shor; it is an older document, and its denial holds for the model, not for the apparatus. Whether the threshold theorem holds physically at scale is **[S]** and is decided not in theory but on the factoring curve.

Table 10: Factoring with Shor's algorithm versus classical records and building blocks (as of September 2026)

Year	Shor on hardware	classical (NFS)	building blocks
2001	$N = 15$	—	—
2012	$N = 21$ (with foreknowledge)	—	—
2019	attempt at $N = 35$ failed	—	—
2020	—	RSA-250 (829 bit)	—
2024	—	—	error correction below threshold (Google Willow)
2025	—	—	48 logical qubits (Quantinuum Helios)
2026	—	—	94 logical qubits $< 10^{-4}$; logical QFT (12 log. qubits)

The larger "quantum records" ($56\,153$, $4\,088\,459$, 2.6×10^{14}, 1.1×10^{12}) concern numbers of special structure (factors differing in a few bits), solved by annealing or QAOA after classical pre-processing; Gutmann and Neuhaus (IACR ePrint 2025/1237) replicated all of them with a 1981 VIC-20, an abacus and a dog. The largest *simulated* Shor run (39 bit) ran classically on 2048 GPUs.

The Shor curve has stood at $N = 21$ since 2012; the classical curve at 829 bit. The yardstick is therefore empirical: **an advantage must become visible on the Shor factoring curve, not on component benchmarks.** That is the corpus position of Doc. 075 — **[X]** no general exponential quantum advantage demonstrated — and it remains valid in September 2026. The theoretical possibility (Shor polynomial; Weyl does not apply; whether another physical obstruction exists is open) remains **[S]**, not [X]; its burden of proof lies with experiment. Resolution floor (G′), counting limit (state/process) and resource limit (Shor) are three different ceilings; for factoring only the third acts today, and it has not been broken.

.11 Summary

Table 11: Results and status

Theorem	Content	Status
A	$\zeta_{T^4/\mathbb{Z}_3}(s) = (1-3^{-s})\zeta(s) = L(s,\chi_0)$	**[B]**
A′	$L(s,\chi_1)$ measures twist-sector asymmetry	**[S]**
B	Euler product hierarchical by $k = \mathrm{ord}_p(3)$	**[B]**
B′	Fractal damping suppresses Euler factors $k \geq 7$	**[B]**
C	Orbifold zeros at $s = 2\pi i k/\ln 3$ on $\mathrm{Re}(s) = 0$	**[B]**
C′	No resonance of Riemann zeros with Z_3 unit	**[B]**
D	Galois character does not separate; classes $= (\mathbb{Z}/26)^*/\langle 3\rangle \cong \mathbb{Z}_4$	**[B]**
D′	Sector pairing $k \leftrightarrow -k$: $f_1 \leftrightarrow f_2$, $f_3 \leftrightarrow f_4$ — 4 classes → 2	**[B]**
D″	Negation couples order 26 to 13; no sub-lepton layer	**[B]**
D‴	$\{f_1, f_2\}$ Majorana layer, $\{f_3, f_4\}$ Dirac layer (sector pairing = Majorana criterion)	**[B]**
D⁗	ν_3 Dirac (O4); m_{ee} from ν_1, ν_2 only (≤14.4 meV); IH more strongly excluded	**[B]**
E	$p = 13$ unique prime with $\mathrm{ord}_p(3) = 3$ worldwide; primes forced by $3^k - 1$	**[B]**
E′	$\mathrm{ord}_p(3) = k$, $3 \mid k \Rightarrow p \equiv 1 \pmod 3$ (Z_3-compatible)	**[B]**
F	Euler factor threshold $p^* \approx 9.76$ $(s = 2)$ = 7-limit boundary	**[B]**
F′	Product density: chance level from $p_{\max} \approx 30$ $(\pm 0.5\,\%)$	**[B]**
F″	Weyl obstruction: limit lies in the object, not the tool (Doc. 316)	**[B]**
G	Coupling: $Q_c = \pi/\sqrt{6\varepsilon} \approx 12.5$ at $1.05\,\%$; limit $k \approx 3$–4 = critical point $\kappa = 1$	**[B]**
G′	Same threefold division, different law: Arnold K^q vs. Farey $1/q^2$	**[B]**
G″	FFGFT dynamically deeply subcritical ($K_{\mathrm{eff}} = 2\pi\xi$); discreteness topological, not dynamical	**[B]**
G‴	Theorem F on $\mathbb{Z}/N\mathbb{Z}$ (RSA) only as analogy	**[S]**
H	$1.05\,\%$ (Doc. 338) $\approx 100\xi$ (Doc. 162): limit $k \approx 3$ is that of one fractal level	**[B]**
H′	At $\varepsilon = \xi$: $Q_c \approx 111$, $p^* \approx 87$, Galois resolved to $k \geq 8$	**[B]**
H″	TFLN $Q > 10^6$ resolves ξ, not ξ^2; ξ = resolution floor, not selector	**[B]**
H‴	Subtractability of the 100ξ level; Doc. 186: locking/Fibonacci/B-meson refuted	**[S]**
I	Resonance factoriser capped at $N_{\max} \approx 1/\varepsilon$ (13–26 bit); classical: counting limit, Landauer erasure only	**[B]**
I′	Circuit model digital (Weyl inapplicable); physical apparatus = resonator, subject to ε/Weyl until threshold theorem demonstrated **[S]**; curve at $N = 21$ since 2012	[X]

The central statement: the full Riemann zeta $\zeta(s)$ belongs to the flat $\mathbb{R}^4$ spectrum. The zeta of the T^4/Z_3 orbifold is $L(s,\chi_0) = (1-3^{-s})\zeta(s)$ — a filtered version in which the 3-Euler factor is separated out and the primes are weighted by the Galois hierarchy $k = \mathrm{ord}_p(3)$. This is the zeta-function form of the entry rule of Doc. 342.

Check scripts

Directory `2/python/Dok343_Skripte/`, wrapper `run_all_343.sh`:

- `pruef_343_zeta_galois.py` — Theorems A–D; partial zeta decomposition, L-functions, zeros, resonance test, weighted Euler product.
- `pruef_343b_klassen_symmetrie.py` — Theorem D; explicit Frobenius classes in $\mathrm{GF}(27)^*$, action of inversion and negation, cosets of $\langle 3\rangle$ in $(\mathbb{Z}/26)^*$, exact values of $L(s,\chi_1)$.
- `pruef_344_primzahlmuster.py` — Theorems E–F; patterns in $\mathrm{ord}_p(3)$, exact threshold p^*, Galois vs. random hit rate, Euler factors.
- `pruef_343c_kopplung_zahlen.py` — Plain-terms II section; Farey coverage and $Q_c(\varepsilon)$, law comparison Arnold K^q vs. Farey $1/q^2$, regime of $K_{\mathrm{eff}} = 2\pi\xi$, Galois coverage by $p_{\max}$.
- `pruef_343f_zweite_schicht.py` — Theorem D‴; classes, orbits, involutions $k \mapsto -k$ and $k \mapsto k^{-1}$, Dirac/Majorana partition.
- `pruef_343d_xi_aufloesung.py` — Theorem G′; threshold table for $\varepsilon \in \{1.05\,\%,\ 100\xi,\ \xi,\ 1/Q_{\mathrm{TFLN}},\ \xi^2\}$.
- `pruef_343e_dok186_check.py` — check of Doc. 186: TFLN quality, Arnold tongues at Fibonacci approximants, 50-GHz ladder, B-meson statistics, winding numbers vs. Fibonacci.

  All assertions passed.

Note on the use of AI assistance

This document was prepared with the assistance of Claude (Anthropic). All numerical computations are verified by Python check scripts. The physical interpretations and conclusions are those of Johann Pascher (ORCID 0009-0000-6518-4064).

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